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Studierichting: Informatica
Oefeningenreeks 7E nr. 3.1

$$f(x, y, z) = 2x^2 - 2xz + 2z^2 + 6xy + 9y^2 - 6z + 9$$

$$\begin{cases} \frac{\partial f}{\partial x} = 4x - 2z + 6y \\ \frac{\partial f}{\partial y} = 6x + 18y \\ \frac{\partial f}{\partial z} = -2x + 4z - 6 \end{cases}$$
$$\Rightarrow y = -1, x = 3, z = 3$$

$$f(x, y, z) = (x - z)^2 + (x + 3y)^2 + (x - 3)^2 \geq 0$$

0 in $(3, -1, 3)$ dus hier is een minimum

References